

7 The southern damselfly, *Coenagrion mercuriale*, is a flying insect found in most of Europe.

Figure 7.1 shows a southern damselfly.



Figure 7.1

(a) Complete Table 7.1 to show the classification of the southern damselfly.

Table 7.1

domain	Eukarya
kingdom	Animalia
phylum	Arthropoda
.....	Insecta
order	Odonata
.....	Coenagrionidae
genus	
species	



- (b) The domain Eukarya includes the kingdoms Animalia and Plantae.

Describe the differences between organisms in Animalia and Plantae.

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- (c) Southern damselflies require areas of open vegetation with slow-flowing water in which to lay their eggs.

- Damselfly eggs hatch into larvae within five weeks.
- Damselfly larvae spend two years underwater before emerging as an adult.
- Damselfly larvae eat other insect larvae and freshwater shrimps.
- Adult damselflies live for only a few weeks.

Southern damselfly numbers are decreasing and this species is listed as endangered in the UK.

- (i) Suggest possible reasons why the southern damselfly has become endangered in the UK.

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- (ii) The mark-release-recapture method has been used to estimate the population size of **adult** southern damselflies.

Suggest why using this method to estimate the population size may be inaccurate.

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[Total: 12]



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